



TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, HTML5, CSS3

Frontend: React, Tailwind CSS, Bootstrap, responsive UI design

Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, MQTT

Databases & Cloud: MongoDB, MySQL, Firebase, AWS

Applied AI / ML: TensorFlow Lite, YOLOv8, MobileNetV2

Embedded / IoT: ESP32, Arduino, Raspberry Pi, Jetson Nano

Tools: Git & GitHub, VS Code, JWT auth, Cursor / Claude Code

TRAINING & CERTIFICATIONS

Full Stack Web Development

SLA, Virtual
Jan 2025 - Jul 2025

- Covered HTML, CSS, JavaScript, React, Node.js, Express.js, and database integration, plus REST APIs and authentication basics.

Diploma in Computer Applications (DCA)

CSC, Nagapattinam
Jan 2021 - Jan 2022

- Covered fundamentals of computer systems, MS Office, basic programming, and internet technologies.

SUMMARY

Full-stack developer building across the MERN stack, currently extending into applied AI and embedded systems as system lead on an Autodesk-sponsored Smart India Hackathon (SIH) robotics project. Six public/private repositories spanning ed-tech, data platforms, security tooling, and AI assistants, built end-to-end from database to UI. Looking for an internship or entry-level role to apply this range on a product team.

FEATURED PROJECT

Sentinel — Autonomous Biomedical Waste Collection Robot | SIH26115, 2026
Autodesk-sponsored

- Serving as system lead on a 6-person team designing an AI-powered mobile robot that navigates hospital wards to collect and segregate biomedical waste.
- Architecting the vision pipeline (YOLOv8n, MobileNetV2, TensorFlow Lite) for waste-type classification on Raspberry Pi / Jetson Nano hardware.
- Specifying the ESP32/Arduino-driven segregation mechanism and MQTT-based telemetry link to a FastAPI backend and React monitoring dashboard.
- Advanced from problem statement to a working concept prototype for the college-level hackathon round, ahead of the national Grand Finale.

PROJECTS

CodeSphere | TypeScript, React

- Built a web-based coding learning and assessment platform for CSE students with C programming practice, timed coding rounds, and leaderboards.
- Implemented automated code evaluation and smart question randomization to prevent repeat exposure across attempts.

AirIndex India | TypeScript, Data Visualization

- Built a real-time airfare price intelligence platform that collects and analyzes flight fare data across major Indian domestic routes.
- Designed an experimental Airfare Price Index with visual dashboards to track fare trends over time.

CO-PO (OBE-AI) | TypeScript, AI Tooling

- Built an AI-powered academic tool that converts uploaded syllabi into measurable Course Outcomes.
- Automated generation of editable CO-PO-PSO mapping matrices with score averaging and a faculty review step.

SecureWipe | TypeScript, Security

- Built a secure data-wiping platform for IT asset recycling with NIST 800-88 and DoD 5220.22-M compliant wipe workflows.
- Generated tamper-proof digital wipe certificates with SHA-256 verification for audit trails.

Zana AI | Python

- Built a personal AI voice assistant supporting natural-language interaction, music control, and voice-driven tool use.

Additional Academic Projects

- Full-stack Student Management System (Node.js, Express, MongoDB/MySQL) with CRUD and auth; College Issue Reporting System (MERN, JWT auth, role-based admin dashboard); E-commerce mini-app (React, Node.js).

EDUCATION

Bachelor of Engineering (B.E.), Computer Science & Engineering | 2024 - 2028
Jeppiaar Engineering College

Diploma in Computer Applications | CSC Computer Education, 2021 - 2022
Mayiladuthurai

LEADERSHIP

Technical Lead | AWS Club, Jeppiaar Engineering College

- Lead technical sessions and hands-on workshops on AWS cloud fundamentals for club members.
- Plan the club's technical roadmap and coordinate cloud-focused events and projects with the core team.

EXTRACURRICULAR ACTIVITIES

- Assisted in organizing and coordinating coding competitions and hackathons at the department level.
- Coordinated a department symposium, managing event planning, scheduling, and team collaboration.